

A close-up photograph of a laboratory microscope with a white body and black eyepiece, positioned next to several glass beakers and a test tube with a red cap. The background is blurred, showing more laboratory equipment.

NEW

2026

Ethiopian Health Profession Licensing Examination(EHPLE)

INFORMATION BOOKLET

MEDICAL LABORATORY SCIENCE



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MINISTRY OF HEALTH-ETHIOPIA

የዜጎች ጤና ለሀገር ብልጽግና!
HEALTHIER CITIZENS FOR PROSPEROUS NATION



AAU-Testing Center

Message From the State Minister, Ministry of Health -Ethiopia



H.E Mrs. Frehiwot Abebe

Improving healthcare quality is a global priority for sustainable development, with high quality healthcare being a key component of universal health coverage. One strategy to maintain health care standards is through provision of health professionals' competency assessment. Consequently, in 2019, the Ministry of Health-Ethiopia, initiated the Ethiopian Health Profession Licensing Examination (EHPLE) for undergraduates in seven health disciplines, which has since expanded to include 17 health disciplines.

The main goal of this competency assessment is to identify health professionals with minimal competencies necessary to perform their duties safely and competently, thus enhancing the quality of health care services. This initiative is overseen by a dedicated Health and Health Related Institutions and Professionals' Regulatory Lead Executive Office (LEO), comprising four desks, which plays a pivotal role in strengthening the system and enabling the LEO to conduct the competency exam more extensively and with improved organization and quality.

It is important to note that this competency assessment differs significantly from traditional academic or employment examinations. Hence, this information booklet has been created to address the informational needs of both examinees and teaching faculty regarding the EHPLE. Additionally, it aims to facilitate the assessment process, while promoting transparency and ensuring the sustainability of the program.

The preparation of this booklet involved the collaboration of esteemed experts from various higher education institutions, AAU-IER, the Ministry of Health, JHPIEGO-Ethiopia, Amref, Health Professionals 'Associations, and the Ministry of Education. Their invaluable contributions are acknowledged with sincere gratitude, alongside appreciation for the Ministry of Health staff for their unwavering commitment and hard work throughout the project.

Acknowledgements

This Information Booklet for EHPLE is a contribution from several educators, researchers, students and concerned individuals with a genuine interest to propel Ethiopia's medical and health sciences education forward.

The Ministry of Health is grateful for the contribution of many individuals and institutions in realizing this endeavor. Among these are Professional Associations, Student Association, Higher Education Institutions (both public and private), JHPIEGO-Ethiopia, AMREF/HWIP, MoE (Ethernet), UNFPA, AAU-IER and all HHRIPR LEOs staff.



Acronyms and Abbreviations

EHPLE	Ethiopian Health Profession Licensing Examination
ETA	Educational and Training Authority
HEIs	Higher Education Institutions
HHrIPR-LEO	Health and Health-related Institutions and Professionals Regulatory Lead Executive Office
HSTP-II	Health Sector Transformational Plan-II
MCQ	Multiple Choice Question
MOH	Ministry of Health
WHO	World Health Organization

Purpose of the Information Booklet

The EHPLE Information Booklet serves as a comprehensive guide for those individuals seeking information about the exam. It typically outlines basic information for candidate registration, exam development and administration processes and procedures, result notification, and the licensing process. It also includes information on the exam framework, i.e., the exam domain, sub-domain, content, process, and task, with sample exam items specific to each profession.

The publication of this Booklet is crucial for the following reasons:

- **Clarity and guidance:** It provides clear information about the exam by ensuring candidates understand the necessary information to prepare them.
- **Accessibility:** It serves as a readily accessible resource for individuals pursuing to take the exam, consolidating essential information in one document and facilitating easy access to necessary details. It also helps other stakeholders who might be interested in such resources.
- **Transparency:** It promotes transparency in the examination process and fosters trust among stakeholders about the exam.

In summary, the publication of this Booklet is essential for creating a transparent, standardized, and accessible framework that guides candidates through the EHPLE process.

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Definition of terms

- **Domain:** a broad category or area of knowledge or skills of a profession
- **Sub-domain:** a subset of a broader domain that focuses on knowledge or skills related to the overarching domain
- **Content:** a more specific subcategory, which is a breakdown of the sub-domain
- **Task:** the responsibility, knowledge, skill, and attitude of a junior undergraduate professional in an actual work environment
- **Process:** a systematic sequence of steps or actions designed to achieve a specific outcome
- **Learning outcome:** a clear and measurable statement that describes what the examinee is expected to know or be able to do
- **Relative emphasis:** the proportional importance or weight assigned to different content areas or categories within the assessment
- **Item:** a particular multiple-choice question
- **Item developer:** a subject matter expert responsible for writing test items or questions that make up the examination
- **Item reviewer:** a subject matter expert responsible for reviewing and refining the test items or questions that make up the examination
- **Standard setting:** a process of determining a cut-off point or passing score for an exam
- **Item difficulty index:** a statistical measure that indicates the proportion of examinees who answered a particular test item correctly
- **Discrimination index:** a statistical measure that evaluates how well a particular test item differentiates between high-performing and low-performing examinees
- **Admission paper:** a printout paper generated by the system after completing registration that contains the examinee's photo, QR code, and necessary information

1. Introduction

1.1. Background

Competency assessment is one of the strategies for controlling the standard of healthcare services provided in healthcare facilities. The World Health Organization (WHO) recommends all healthcare professionals to have necessary competencies. In Ethiopia, the Health Sector Transformational Plan-II (HSTP-II) states competency assessment of all graduates before joining the health workforce as one of the strategic initiatives.

The Ministry of Health (MOH) launched the Ethiopian Health Profession Licensing Examination (EHPLE) for undergraduates in 2019. The Health and Health-related Institutions and Professionals Regulatory Lead Executive Office (HHrIPR-LEO) of the Ministry of Health is given a mission to implement the Ministry's objective to achieve competency-related goals. It has the responsibility to ensure that the EHPLE meets technical, professional, and legal standards, and to protect the health, safety, and welfare of the public by assessing candidates' abilities to practice competently.

Currently, the exam is given for 17 health professions (Medicine, Nursing, Public Health, Pharmacy, Medical Laboratory Science, Anesthesia, Midwifery, Dental Medicine, Medical Radiology Technology, Environmental Health, Psychiatric Nursing, Pediatric and Child Health Nursing, Emergency and Critical Care Nursing, Human Nutrition, Surgical Nursing, Optometry, and Physiotherapy). Since its introduction until December 2025, a total of 204,811 examinees took the exam in 23 rounds.

1.2. The Rationale of EHPLE

One of the critical functions of the MoH is to guarantee the efficiency, quality, and equity of healthcare delivery and to protect the public from any undesirable consequences in healthcare delivery practices. As professionals' competence is a significant determinant of the quality of health, evaluation of health professionals' competence has now been given due attention. The licensing examination for health professionals serves as a crucial step to ensure that individuals entering the field meet specific competency standards. The sole aim of the competency assessment is to safeguard public health by verifying that health professionals have the minimal basic knowledge, attitude, and skill required to provide safe and effective care.

Licensing exams act as a preventive measure, ensuring that only competent professionals join the health workforce, which, in turn, contributes to reducing the occurrence of medical errors and enhancing overall patient safety. By setting standards through examinations, regulatory bodies strive to minimize the risk of medical errors caused by incompetence.

2. Key processes of EHPLE

EHPLE involves several key processes to ensure the quality and reliability of the examination.

2.1.Registration of candidates

EHPLE has a mandatory online registration system for both new and repeat candidates, which can be found at <http://hple.moh.gov.et>

Please note these important notes during registration.

New Test Takers:



- The list of eligible candidates from governmental and private Higher Education Institutions (HEIs) will be uploaded to MoH's online registration system by the registrar office of their respective HEIs.
- Candidates who completed their education abroad must bring the following documents in person to MoH in order for their names to be uploaded to the online registration system:
 - ✓ Equivalence document from ETA
 - ✓ Externship attachment completion letter after completing the externship assigned by the regulatory body
- Once the name of the candidate is uploaded to the system and registration has opened for the current exam round, the candidate must register at <http://hple.moh.gov.et> by uploading the necessary documents listed below.
 - ✓ a scanned original or temporary degree
 - ✓ a scanned national ID

Repeat Test Takers:



- Since the information about re-exam candidates already exists in the system, the candidate should register by directly going to <http://hple.moh.gov.et>. There is no need for re-exam candidates to upload their documents.

Both new and repeat candidates:



- After completing the registration, the candidate must download and print the admission paper by logging into his/her account using username and password, or national ID FAN.
- The candidate can change the exam center by logging into his/her account only during the registration period
- An application for center change is not allowed after the registration period is closed.

2.2.Task Analysis

The first step of exam development involves conducting a comprehensive task analysis study, which identifies the tasks, knowledge, skills, and abilities required from a junior undergraduate professional in the specific profession. The analysis is typically done through surveys, interviews, or observations of practitioners in the actual work environment, as well as through the Delphi method with subject matter experts.

2.3.Exam Blueprint

Based on the task analysis findings, the exam blueprint is prepared that outlines the content areas to be covered in the examination and the weight or emphasis given to each area. This ensures that the exam reflects the key competencies and knowledge needed for competent practice in that specific profession. Blueprint is the matrix or chart that shows the number and type of test questions represented across the topics in the content area, consistent with the learning outcome and relative weight of the test given to each content area. The blueprint also identifies the percentage weighting of cognitive dimensions as the level of competence tested in each knowledge domain.

Key components of blueprint are:

- Domain
- Sub-domain
- Content
- Task
- Process
- Learning outcome
- Assessment methods
- Assessment tools/instrument (test format)
- Relative emphasis (in percentage)

2.4.Item Development

The items are developed following specific guidelines to ensure clarity, relevance, and fairness. Subject matter experts with experience in the field are selected from HEIs to develop exam questions (items) that align with the exam blueprint. The exam questions focus mainly on “knows how” according to the competency level of the Miller's pyramid. The items are produced in a secure location on designated computers that are free from internet connectivity. The items are scenario-based and constructed with stem, lead-in, and four options/alternatives. All items will have a single-best-answer type of Multiple-Choice Question (MCQ) that addresses the learning outcome defined in each content area. Standard textbooks, updated guidelines, and standards are used as reference materials.

2.5.Item Review

Once developed, the items undergo a rigorous review process by item reviewers. The main purpose of the exam review process is to evaluate content relevance, technical accuracy, clarity, and sensitivity related to culture and religion. More experienced subject matter experts as well as psychometric experts will do the review to ensure the items meet psychometric standards. Subject matter experts shall review the items to confirm that they are accurate, clearly stated, and correctly keyed using the checklist. Psychometric experts shall review the items to ensure that

they are not technically flawed. They also work on editorial review to check grammar, punctuation, and spelling errors. This helps ensure the reliability and validity of the items.

2.6. Standard setting method

The standard setting or cut-off point of the EHPLE is determined using the Modified-Angoff method, which is one of the most widely used and legally defensible standard setting approaches to set a cut-off point for high-stake competency examinations.

The method involves a panel of subject matter experts who evaluate each exam question and then estimate the probability that a minimally competent examinee would answer each test item correctly. The average of the experts' predictions for an exam question becomes its predicted difficulty. The average of the predicted difficulty values across all items on a test is the recommended cut-off point. This point indicates the minimum level of knowledge and skill required to pass.

2.7. Exam Administration

The EHPLE is administered following established protocols and guidelines. Proper exam administration procedures, appropriate security measures, and appropriate consideration for test-takers who need special support will be applied during exam administration at exam centers. The exam is administered in selected HEIs nationally, where candidates can choose based on their convenience at the time of registration. The exam schedule will be posted ahead of time on the MoH website and social media pages. Examinees who have a valid admission paper are eligible to sit for the exam. The mode of exam administration is computer-based testing.

CAUTIONS

-
- Candidates are allowed



- Attend the orientation session in order to sit for the exam
 - Arrive at the exam center on time
 - Bring a legal ID and admission paper
 - Complete the exam within the allotted time frame
-

- Candidates are **NOT** allowed



- To bring reference materials, blank paper, or notes into the exam center
- To smoke, eat, or drink in the exam room
- To bring mobile phones, tablets, smart watches, camera devices, eyeglasses, calculators or any type of electronic device into the exam center
- To bring their personal belongings to the exam center
- To bring weapons and sharp materials into the exam center
- To give or receive assistance to or from other candidates during the examination

2.8. Scoring and post exam analysis

Once the exam is completed, the scoring process begins. The exam scoring process involves computerized scoring using software.

Post-exam analysis is the process of analyzing examinees' responses to individual test items in order to assess the quality of the items and the exam as a whole. This phase helps to identify any poorly performing items that may need revision or removal from the exam. The item difficulty index, discrimination index, and reliability coefficient are elements of exam analysis.

2.9. Result notification and appeal management

After scoring and analysis, individual score reports are generated and provided to examinees through the website <http://hple.moh.gov.et>. After result notification, examinees can submit their appeal through phone or email within 10 working days after result notification.

2.10. Licensing

The list of examinees who passed the exam will be sent to regional and city administration regulatory bodies. A license is obtained from the regional/zonal health bodies where he/she permanently lives.

Requirements for professional licensing are:



- Passing the EHPLE
- Original or temporary degree
- Educational documents (10th and 12th certificates)
- Medical certificate
- Government issued ID
- Additional prerequisites based on the requirements of regional regulatory bodies

3. Exam framework

The key broader professional roles, also known as domains or main knowledge areas serve as a building framework for the licensing examination content for medical laboratory science professionals. The domains are further divided into discrete professional attributes that constitute sub-units (also referred to as sub-domains) defining the professional identity of medical laboratory science professionals. Tasks specifying the performance level of each sub-domain serve as the final characteristic of the professional duties on which the licensing exam focuses.

The contents of the licensing examination are presented below, structured into key roles (domains), sub-units (sub-domains), and tasks. The examination emphasis for each domain and sub-domain, out of the total 100% questions, is indicated in brackets.

Key professional roles/ domains

- ☐ Patient Care (81.0%)
- ☐ Laboratory Quality Assurance and Safety care (6.0%)
- ☐ Scholar (2.0%)
- ☐ Professionalism (3.0%)
- ☐ Leadership and management (3.0%)
- ☐ Health promotion and disease prevention (5.0%)

Key role/ domain 1: Patient care (81.0%)

Description: This domain encompasses the professional roles of laboratory technologists in the provision of high-quality, safe, and patient-centered laboratory service within their scope of practice. The provision of up-to-date, ethical, and resource-efficient laboratory service requires the application of integrated knowledge of biomedical, clinical, behavioral, and social sciences. This is conducted in collaboration with patients and their families, other healthcare professionals, and the community. As patient care providers, laboratory technologists shall implement the pre-, analytic, and post-analytic phases of laboratory services. These phases include receiving test requests, identifying patients and specimens, collecting, transporting, accessioning, processing, testing, reporting test results, interpreting, following up, storing, and retesting as necessary. To demonstrate competence in this domain candidates shall apply such integrated knowledge in the following sub-areas:

- ☐ Clinical Chemistry (10.0%)
- ☐ Medical virology (5.0%)
- ☐ Mycology (4.0%)
- ☐ Bacteriology (10.0%)
- ☐ Molecular Biology (4.0%)
- ☐ Urine and Body Fluids Analysis (6.0%)
- ☐ Hematology (10.0%)
- ☐ Immunohematology (6.0%)

- ☐ Serology (6.0%)
- ☐ Medical Parasitology (11.0%)
- ☐ Instrumentation (2.0%)
- ☐ Introduction to Medical Laboratory Sciences (4.0%)
- ☐ Histopathology (3.0%)

Key role/ domain 2: Laboratory Quality Assurance and Safety care (6.0%)

Description: This domain encompasses the professional roles of laboratory technologists in quality assurance and clinical safety activities to promote public health and to ensure that clinical laboratory practice is safe, effective, good quality, and as per good practices. Implementing an up-to-date, ethical, and resource-efficient quality assurance system requires the application of integrated knowledge of GMP, biomedical, clinical, behavioral, and social sciences. As quality assurance focal persons, laboratory technologists shall perform routine quality control inspections, troubleshoot technical issues, comply with quality management system guidelines, and actively engage in quality improvement initiatives. Laboratory technologists shall also possess essential clinical safety applicable knowledge including adherence to safety protocols, implementation of equipment safety measures, ensuring emergency preparedness, and implementing environmental safety measures. To demonstrate competence in this domain candidates shall apply such integrated knowledge in the following sub-areas:

- ☐ Quality assurance (5.0%)
- ☐ Safety in clinical laboratory (1.0%)

Key role/ domain 3: Scholar (2.0%)

Description: This domain encompasses the professional roles of medical laboratory science professionals in generating and utilizing scientific data to improve the health and well-being of Ethiopians and broaden their scientific knowledge within the healthcare system and community setting. Providing this service requires the application of integrated knowledge in research methods, measurements of health and disease, biostatistics, epidemiology, clinical audit, evidence-based practice, and research ethics. To demonstrate competence in this domain, candidates must possess applied knowledge in planning, problem identification, data collection, analysis, interpretation, report writeup, and dissemination of research outputs.

Key role/ domain 4: Professionalism (3.0%)

Description: This domain encompasses the professional commitment of medical laboratory science professionals to promoting the health and well-being of individuals and society through

adhering to ethical standards, maintaining personal integrity, and upholding high standards of competence in all areas of practice. To exhibit competence in this domain, candidates must possess applied knowledge of ethical principles, medico-legal practices, effective communication, accountability to the profession and society, maintenance of professional excellence and personal health, and professional values such as compassion, respect, integrity, honesty, altruism, and humility.

Key role/ domain 5: Leadership and management (3.0%)

Description: This domain encompasses the professional roles of medical laboratory science professionals in envisioning a high-quality healthcare system through self-awareness, active participation in healthcare teams, leading teams, and managing health systems. Providing this service requires the application of integrated knowledge in continuous quality improvement, effective health system leadership, management, and healthcare ethics.

Key role/ domain 6: Health promotion and disease prevention (5.0%)

Description: This domain encompasses the professional roles of medical laboratory science professionals in enhancing the health and well-being of patients, communities, and the larger populations they serve through health advocacy, disease prevention, health promotion, health protection, and the promotion of health equity. Providing this service takes an integrated understanding of determinants of health, health informatics, epidemiology, communicable disease control, and health education.

Table 1 Exam Content for Medical Laboratory Science Profession

Domian: Patient care
Sub-Domian: Clinical Chemistry
Task
Collect, handle and process blood sample properly
Prepare and identify the Solutions
Apply the basic principles of Chemistry Analyzers
Identify assay technique used for each clinical chemistry parameter
Measure Glucose Concentration from body fluids
Determine Lipid Profile
Determine Renal Function Test
Perform Liver Function Test
Perform cardiac Function Test
Perform electrolyte Test
Perform Diagnostic enzymology test
Determine Hormone analysis
Perform Tumor marker analysis
Ensure Quality Assurance in Clinical Chemistry
Sub-Domain Medical Virology
Task
Collect, transport and preserve specimens
Perform Diagnostic methods in Virology
Identify Medically important DNA Viruses
Differentiate ds RNA virus
Distinguish Positive sense ss RNA
Differentiate Negative sense ss RNA
Differentiate RNA-RT (Retrovirus)
Identify Emerging and re-emerging viral diseases
Ensure Quality assurance in Medical Virology
Sub-Domin: Mycology
Task
Collect, transport, process and preserve Mycology specimen
Perform Fungal Diagnosis
Diagnose Superficial mycosis
Diagnose Cutaneous mycoses
Diagnose Subcutaneous mycosis
Diagnose Systemic mycosis
Cary out mychological test for the diagnosis of Opportunistic mycosis
Identify MYCOTOXINS AND MYCOTOXICOSES

Sub-Domain:- Bacteriology

Task

Prepare Bacteriological test reagents
 Perform Bacteriological staining
 Collect, Transport and Process Bacteriological specimens
 Store and dispose bacteriological wastes
 Prepare Culture media
 Perform appropriate Biochemical Test for identification of Micro-Organisms
 Perform Antimicrobial susceptibility test
 Identify Pathogenic Gram positive cocci
 Differentiate Gram-positive rods
 Identify Gram negative cocci
 identify Gram negative coccobacilli
 Identify Gram negative rods
 Identify Mycobacterium spp.
 Identify Obligate intracellular bacteria
 Identify Spirochete

Ensure Quality assurance in Bacteriology

Sub-domain: Molecular Biology

Task

Perform nucleic acid Extraction
 Perform nucleic acid Purification
 Perform nucleic acid Amplification
 perform Gel electrophoresis
 Perform Blotting techniques (Northern and Southern blotting)
 Ensure Quality Assurance in Molecular Biology

Sub-Domain: Urine and Body Fluids Analysis

Task

Collect, preserve, transport process urine and other body Fluid Specimen
 Perform Physical examination of urine and other body Fluid specimen
 Perform Chemical examination of urine and other body fluids
 Perform Microscopic examination of Urine and Other body fluids Specimen

Sub-Domain: Hematology

Task

Collect Blood specimen
 Prepare thick and thin Smear
 Perform Hematological Staining
 Perform Hemocytometry and differential cell count from blood and body fluids
 Perform Hemoglobin ,Hematocrit, ESR measurement Osmotic Fragility tests and determine Red cell indices.
 Examine Red cell morphology
 Diagnose and Identify types of Anemia
 Identify Leukocyte disorders

Perform Coagulation Profiles (PT, APTT, INR)
Prepare, stain and examine BM smears and LE cell
Operate Hematological Automations
Ensure Quality Assurance in Hematology
Sub-Domain;-Immunoheamatology
Task
Perform Immunoheamatological tests
Sub Domain:-Serology
Task
Perform Agglutination tests
Perform Immunochromatograph test
Perform Enzymeimmuno assay(ELISA) test
Perform Fluorescent antibody tests (FAT)
Operate Flowcytometer
Ensure Quality in Serology
Sub-Domin:- Medical Parasitology
Task
Detect and Identify Intestinal Nematodes
Identify Blood and tissue nematodes
Identify Cestode
Identify Trematode
Identify Parasitic Amoebae
Free living amoebae
Identify Luminal (Urogenital & Intestinal) flagellate
Identify Tissue and blood flagellate
Identify intestinal coccidian
Identify Blood & tissue coccidians
Identify Balantidium coli
Ensure Quality assurance in Medical Parasitology
Sub-Domain:- Instrumentation
Task
Operate Water purification and sterilization instruments and biological safety cabinet, Incubators and ovens
Operate Automated micropipettes
Operate Spectrophotometer
Operate Electrochemical instrument
Sub-Domain:- Introduction to Medical Laboratory Sciences
Task
Identify, handle and Use Laboratory glassware and plasticwares
Identify, handle and use Laboratory equipment
Prepare and store Laboratory reagent and solution
Collect, handle and transport Clinical Specimen
Perform Sterilization and disinfection

Apply Laboratory safety
Sub-Domain:- Histopathology
Task
handle, and Fix /preserve Histopathology specimen
perform Tissue processing
Perform Tissue Sectioning
Prepare Staining reagents
Apply Special staining methods
Domain: Laboratory Quality Assurance and Safety care
Subdomain:- Quality Assurance
Task
Identify errors in clinical laboratories
Perform values of diagnostic test
Perform verification of and/or validate test methods using quality control materials
Manage Specimen, and Prepare and implement standard operating procedures
Operate laboratory instrument and equipment
Run evaluation and interpret quality control samples/materials
Describe laboratory analytical method using absurd value check, duplicate analysis, Delta check using patient samples as quality control
Determine of Reference Interval
Assess External Quality
Participate in laboratory accreditation process
Subdomain:- Safety in clinical Laboratory
Task
Create safe laboratory rooms and surrounding working environment
Apply safety precautions during laboratory process/activities
Domain: Professionalism
Task
Adhere to the ethical principles to medical laboratory practice
Apply legal issues to govern professional practice
Provide compassionate, respectful service
Communicate, advocate and collaborate effectively
Domain:- SCHOLAR
Task
Analyze data and interpret Research result
Formulate and Summarize research result and measurement
Domain:- Leadership and Management
Task
Apply Leadership, Health system management ,Health laboratory management
Organize laboratory activities based on the plan

Monitor & Evaluate Activities

Domain:- Health promotion & Disease prevention
Task
Apply disease prevention and health promotion
Investigate Outbreaks
Prepare action plan
Analyze food microbials & preserve & store Foods
Analyze water and beverage microbials
Examine food borne pathogens
Identify microbial indicators for food safety and quality
Assess Bioterrorism
Identify medically important Vectors, their characteristics, mode of disease transmission and control mechanism



Sample questions

1. Blood sample was collected from a 32-year-old female patient for renal function tests measurement. A laboratory technologist processed and analyzed the sample using spectrophotometric method. The results showed high blood urea concentration with normal creatinine and uric acid concentrations.

What is the most likely cause of the elevated urea concentration?

- | | |
|-------------------------------------|---|
| (A) High intake of dietary lipids | (C) High intake of protein rich diet |
| (B) High intake of purine rich diet | (D) High intake of carbohydrate rich diet |

Answer key: The answer is C

Explanation: Urea is the major nitrogen-containing metabolic product of protein catabolism in humans accounting for more than 75% of the non-protein nitrogen eventually excreted. Taking diets rich in protein causes elevation of blood urea concentration. Therefore, the best answer is option C.

2. A laboratory technologist working in a Clinical Chemistry Laboratory wants to measure the glucose level of a 77-year-old female patient and received her blood specimen. The sample was processed and analyzed using spectrophotometric method. The reaction was linear and 80 mg/dl glucose standard solution was used. The absorbance of the standard was 0.2 and the absorbance of the patient sample was 0.4.

What is the most likely concentration of glucose in the patient's blood sample?

- | | |
|--------------|---------------|
| (A) 40 mg/dl | (C) 100 mg/dl |
| (B) 80 mg/dl | (D) 160 mg/dl |

Answer key: The answer is D

Explanation: For calculating glucose concentration in patient sample, we use beer's law

using the formula;
$$\frac{C_u}{C_s} = \frac{A_u}{A_s}$$
 where C_u is concentration of substance in the patient sample which is unknown, A_u is absorbance of unknown, C_s is concentration of standard and A_s is absorbance of standard. Thus, $C_u = 0.4/0.2 \times 80 \text{ mg/dl} = 160 \text{ mg/dl}$. Therefore, the best answer is option D.

3. A laboratory technologist who is working in a urinalysis laboratory received urine specimen of a 45-year-old female patient. When the technologist examined the specimen, it has fruity odor.

What is the most appropriate test that the technologist should perform next?

- | | | | |
|-------------|-------------|----------------|-------------------|
| (A) Nitrate | (B) Protein | (C) Creatinine | (D) Ketone bodies |
|-------------|-------------|----------------|-------------------|

Answer key: The answer is **D**

Explanation: The main purpose of physical examination of urine is to give hints for Medical Laboratory professionals for subsequent examination. The results of the physical portion of the urinalysis can also be used to confirm or to explain findings in the chemical and microscopic areas of urinalysis. One of the physical examination of urine is odor. One of the causes of unusual odor is ketone bodies which produce a sweet or fruity odor. Therefore, option D is the answer for this question.

4. A Medical Laboratory Scientist wants to prepare quality control materials for ensuring validity of glucose measurement. He used leftover serum specimen from the patients visiting Laboratory units and prepared quality control material from serum with the concentration near the level of end of detection.

What is the most likely quality control material prepared by the Scientist?

- | | |
|---------------------|--------------------|
| (A) Critically high | (C) Normal range |
| (B) Abnormally low | (D) Critically Low |

Answer key: The answer is **D**

Explanation: There are different levels of quality control materials in clinical chemistry Laboratory. These include Normal, abnormally high and abnormally low, and critically high and critically low. The Critical low quality control material is a type of quality control used for checking low concentration of substance near the low end of detection. Therefore, the answer is option D.

5. A physician requested the preparation of coagulation profile for a patient with bleeding disorder. The nurse collected the blood sample using EDTA vacutainer tube and sent it to a laboratory for coagulation analysis. However, the laboratory technologist reported that the collected blood is not appropriate for coagulation analysis.

What is the most likely cause for rejecting the sample?

- (A) Specimen was collected at the wrong time
- (B) Specimen was contaminated with intravenous fluid
- (C) Specimen was collected in the wrong anticoagulant
- (D) Test order requisition and the tube identification do not match

Answer key: The answer is **C**

Explanation: Blood sample collected by EDTA vacutainer tube is not suitable for coagulation tests. It is because, calcium which is one of the important component in the blood clotting system, will be removed from the whole blood during anti-clotting reactions by EDTA. Thus, blood collected by EDTA will give false results in the coagulation profile tests.

6. A physician working in a hospital requested a red cell indices values for an anaemic male patient with haemoglobin value of 12.5g/dl. A laboratory technologist reported the red cell indices values as follows: *MCV: 80fl(85fl); MCH, 30pg and MCHC 34g/dl*.

What is the most likely type of anemia in this patient?

- | | |
|------------------------------------|------------------------------------|
| (A) Microcytic hypochromic anemia | (C) Macrocytic normochromic anemia |
| (B) Normocytic normochromic anemia | (D) Macrocytic hyperchromic anemia |

Answer key: The answer is **B**

Explanation: The normal concentration of hemoglobin for a male adult is 13.0-18.0 g/l, and for adult women is 12.0-15.0 g/l. The patient's hemoglobin value of 12.5 indicates slightly anemic condition. In the red cell indices measurement, the MCV, MCH, MCHC values are used for the assessment of red blood cells average volume, weight, and hemoglobin concentration with reference range of 92 +/-9 fl, 29.5 +/-2.5 pg, and 30 +/-15g/l respectively. Accordingly, the patient's RBC indices; MCV: 85 fl; MCH: 30pg, and MCHC: 34g/dl all are in normal ranges, and indicate the normocytic normochromic anemia state.

7. A physician sent a request of Rh-typing for pregnant woman prior to her CS surgery. A laboratory technologist took a blood sample from the woman and made the entire required reagents ready for cell typing.

What is the most appropriate type of anti-serum to be used in this case?

- | | | | |
|------------|------------|------------|-------------|
| (A) Anti-A | (B) Anti-B | (C) Anti-D | (D) Anti-AB |
|------------|------------|------------|-------------|

Answer key: The answer is **C**

Explanation: In the standardized ABO –Rh blood grouping, anti-human globulin, Anti-A, Anti-B, and Anti-D are used for individual's blood typing by conducting agglutination reactions. Anti-A reacts with blood group A, anti B reacts with blood group B, and individuals with blood type AB react with both anti-human globulins A and B. If anti A, and B antihuman globulins make no reaction with individual blood, it is interpreted as "O" blood type. Individual's blood cells that react with anti D and agglutinates are interpreted as Rh positive blood type.

8. In a hospital surgical department, a nurse brings a laboratory request paper for blood cross match to a blood bank laboratory, for a patient who is scheduled for major surgery in the next day. Accordingly, a laboratory technologist going to the ward collected blood sample from the patient who has surgery procedure by following SOP. Then the technologist brings the sample into laboratory doing ABO blood grouping and identifies patient blood type. Then the technologist brings appropriate donors blood bag from the fridge, to do major cross match by doing an agglutination reactions.

What is the most appropriate method to perform the stated major cross-match in this case?

- (A) Mix the patient's serum with the donors blood cells at room temperature
- (B) Mix the patient's red blood cells, with the donor serum at room temperature
- (C) Mix the patient's serum, with the donors blood cells at 37°C temperature
- (D) Mix the patient's red blood cells, with the donor serum at 37°C temperature

Answer key: The answer is A

Explanation: In the transfusion medicine, the two major transfusion reactions are: major incompatibility reactions, and minor incompatibility reactions. During the major incompatibility reactions, antibodies from the recipients will kill donor's red cells; so the technologist tests it by interacting the recipient's serum with the donor's red cells agglutination reactions at room temperature. On the other hand, in the minor incompatibility reactions, the donor's antibodies destroy the recipient's red blood cells. And to diagnose it, the laboratory technologist tests it by interacting the recipient's red cells with the donor's serum agglutination reactions at room temperature.

9. A patient in gastroenterology center was allowed to ingest ^{14}C radio-labeled urea. The carbon dioxide released by urease enzyme of the suspected pathogenic bacteria was detected in the patient's breath using a mass spectrometer.

What is the most likely bacterium diagnosed by the test method?

- (A) Campylobacters jejuni
- (B) Klebsiellae pneumonia
- (C) Helicobacter pylori
- (D) Pseudomonas aeruginosa

Answer key: The answer is C

Explanation: In this scenario, urease breath test used as the screening tests for *H. pylori*. This bacterium is known for causing inflammation, gastritis, peptic ulcers, and gastric cancer. The organism produces urease enzyme that is used as one of the diagnostic biochemical test. This enzyme breaks urea into carbon dioxide and ammonia. The produced carbon dioxide can be detected and used as diagnostic if the urea is radio-labeled. The ^{14}C radio-labeled urea is specifically used to differentiate microbial urease enzyme from carbon dioxide produced during respiration.

10. A mother brought an eight-month-old infant to a pediatric OPD with symptoms of teeth and bone malformation, and widespread desquamating maculopapular rash. The physician suspected syphilis and requested for serological test. The result showed positive for *Treponema pallidum*.

What is the most likely mode of transmission for the pathogen?

- (A) Congenital (C) Aerosol inhalation
(B) Mosquito bites (D) Fecal-oral route

Answer key: The answer is A

Explanation: In this scenario, *Treponema pallidum* is a well known etiological agent for congenital syphilis. The pathogen can be transmitted by sexual contact or congenitally. The probability of getting congenital infection is high if the mother of the infant has history of syphilis case. *Treponema pallidum* has the ability to cross placenta and infect the fetus. After the birth of the infant, the bacteria caused teeth and bone malformation. Whereas, the other modes of transmission listed in the options are not known means for *T. pallidum* transmission.

11. A 26-year-old male patient came to a health center with complaints of high grade fever, arthralgia, shivering and headache. A laboratory technologist processed and examined the blood film using an oil immersion objective and found small rings and some double chromatin dots on the rings. The technologist also observed accolé forms of rings lying on the red blood cell membranes.

What is the most likely plasmodium species seen under the microscope?

- (A) *Plasmodium vivax* (C) *Plasmodium malariae*
(B) *Plasmodium ovale* (D) *Plasmodium falciparum*

Answer key: The answer is D

Explanation: Among different species of plasmodium, *P. falciparum* is the most frequently reported malaria case. This parasite can be diagnosed using Romanowsky stained blood film. In the stained blood film: trophozoites, schizonts, and gametocytes stage of the parasite can be observed. Typical differentiating features of the *P. falciparum* trophozoites stage from other malaria species are having small rings, double chromatin dots on the rings, having accolé forms, and double infection of a single red blood cell (See the picture below).

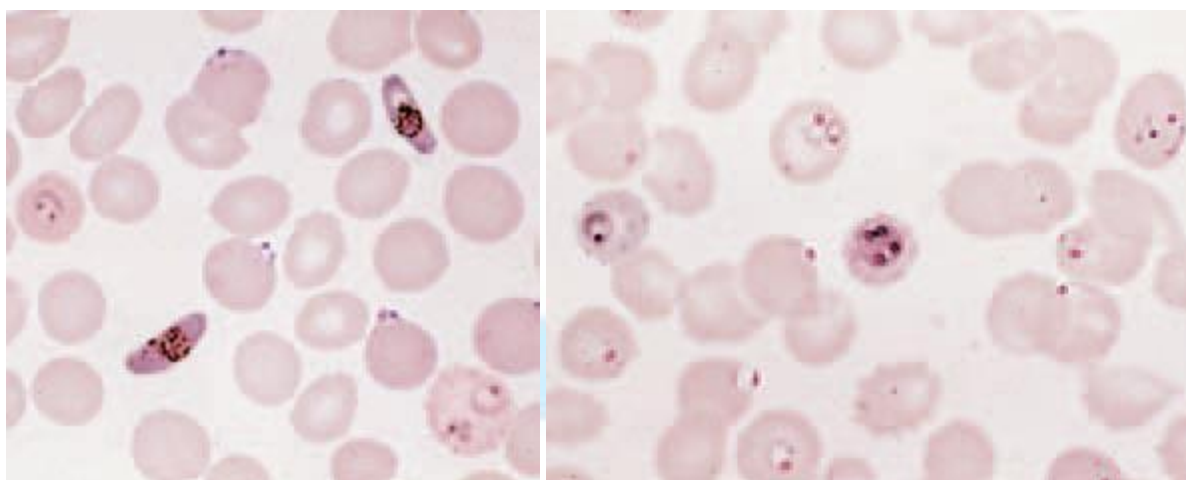


Figure1. *Plasmodium falciparum* trophozoites stage having small rings with double chromatin dots on the rings and accolé forms (Source: Cheesbrough, 2009).

12. Smoking cessation programs, focused on individuals, provide an excellent example of a high-risk strategy and are appropriate since most smokers wish to abandon the habit. Thus, individual smokers and the concerned physicians are usually motivated. Such programs are more likely to be effective when complemented by population approaches to tobacco control.

What is the most likely level of disease prevention to be considered in this intervention?

- | | |
|---------------------------|--------------------------|
| (A) Primordial prevention | (C) Secondary prevention |
| (B) Primary prevention | (D) Tertiary prevention |

Answer key: The answer is **B**

Explanation: This question asks about level of preventions corresponding to different phases of disease development. Among the existing level of prevention, primordial and primary prevention mostly contribute to the health and well-being of the whole population. The question specifically asks about prevention of smoking which is a known risk of diseases. The purpose of primary prevention is to limit the incidence of a disease by controlling causes and risk of factors. Therefore, primary prevention is used for such specific risk factors.

13. A senior laboratory microbiologist, who has been serving in a microbiology laboratory for the last six years, carries out his duty according to professional standards and requirements. In addition, the microbiologist strives to strengthen the professional association and tries to promote the advancement of laboratory tests especially in the area of microbiology.

What is the most appropriate type of responsibility that concerns this microbiologist?

- | | |
|-------------------------|-----------------------------|
| (A) Duty to self | (C) Duty to the profession |
| (B) Duty to the patient | (D) Duty to the environment |

Answer key: The answer is **C**

Explanation: Medical laboratory professionals have various duties and obligation mentioned in Ethiopian Medical Laboratory Association (EMLA) code of ethics booklet. The duty to the profession (option C) can be expressed through: promoting the image and status of the profession by maintaining high standards in the professional practice; using standard of Operating procedure (SOP) for routine activity; and actively participating in professional matters through membership of EMLA. On the other hand, duty to self (option A) is professional commitment to self growth, and engagement of Continuous professional development. In the same manner, duty to patient (option B) explains about duty to ensure the best interests or well-being of patients as a primary professional duty. Duty to environment (option D) explains about the duty to protect environment through proper waste management.

14. A laboratory planned to compare the efficacy of drug „A“ with the efficacy of drug „B“ on randomly selected 100 study subjects. Then, the technologist assigned the study subjects in to two independent groups randomly and treated them with either of the two drugs and followed their progress over certain period of time to see the outcome.

What is the most likely study design used to identify the efficacious drug in this case?

- | | |
|-------------------------------|----------------------------------|
| (A) Cohort study design | (C) Experimental study design |
| (B) Case control study design | (D) Cross-sectional study design |

Answer key: The answer is C

Explanation: In the experimental study design, the researcher is able to manipulate the exposure status to the study participants. In the scenario, the laboratory technologist wants to know the efficacy of drug „A“ and „B“. So the technologist assigned the participants in to two groups and provided the drug for the two groups which shows the manipulation of the researcher on the exposure status of the participants. The other options are completely observational type that means the researcher simply observes the participants without any manipulation on the exposure assignment.

15. A male laboratory technologist working in a health center wanted to have a discussion with the director about the benefit he should get for his extra working hours including getting incentives and promotion. However, the technologist couldn't get the expected incentives and promotion due to the unwillingness of the director for discussion. This situation created a conflict between them and affected the performance of the health centre.

What is the most likely conflict management strategy exercised in this case?

- | | |
|---------------|--------------------|
| (A) Lose-lose | (C) Win-win |
| (B) Lose-win | (D) No-win no-loss |

Answer key: The answer is A

Explanation: In conflict situations, individuals can either suppress conflict or engage in activity, which will lead to its resolution. Behaviors directed toward the resolution of conflict can be characterized by three different conflict resolution strategies, i.e. win- lose, lose- lose, and win - win. Basically, Lose –Lose conflict resolution strategy is an undesirable exercise as most people do not intentionally want to forfeit their interests. However, it becomes an option when other strategies do not work well or when conflicts are not managed on time. In this scenario, the director was not willing for discussion, and the technologist could not get incentives and promotion. So, it was a failure for the director as the performance of the health centre.



Sample References

- Burtis, C. A., & Bruns, D. E. (2015). Tietz Fundamentals of Clinical Chemistry and Molecular Diagnostics (7th ed.). Elsevier.
- Strasinger, S. K., & Di Lorenzo, M. S. (2008). Urinalysis and Body Fluids (5th ed.). F.A. Davis Company.
- World Health Organization. (2011). Laboratory Quality Management System Handbook.
- Yared, A., et al. (2016). Hematology Lecture Notes for Medical Laboratory Students.
- Daniels, G., & Bromilow, I. (2007). Essential Guide to Blood Groups. Blackwell Publishing.
- Cheesbrough, M. (2006). District Laboratory Practice in Tropical Countries, Part 2 (2nd ed.). Cambridge University Press.
- Murray, P. R., Rosenthal, K. S., & Pfaller, M. A. (2012). Medical Microbiology (7th ed.). ELSEVIER.
- Cheesbrough, M. (2006). District Laboratory Practice in Tropical Countries, Part 2 (2nd ed.). Cambridge University Press.
- Cheesbrough, M. (2009). District Laboratory Practice in Tropical Countries, Vol I (2nd ed.). Cambridge University Press.
- Apurba, S. S., & Sandhya, B. K. (2014). Essentials of Medical Parasitology. Jaypee Brothers Medical Publishers (P) LTD. (pp. 90-114).
- Beaglehole, R., Bonita, R., & Kjellstrom, T. (2006). Basic Epidemiology (2nd ed.). WHO Library Cataloguing in Publication Data.
- Webb, P., & Bain, C. (2011). Essential Epidemiology: An Introduction for Students and Health Professionals (2nd ed.). Cambridge University Press.
- Ethiopian Science and Technology Commission, Ethiopian Public Health Association, & Regional State Health Bureaus. (2005). Health Research Method Training Module2.
- Susan, C., Nigel, U., & Tanja, Pless-M. (2007). An Introduction to Public Health and Epidemiology (2nd ed.).
- Cherie, A., & Gebrekidan, B. (2005). Nursing Leadership and Management: Lecture Notes for Nursing.
- Jones, R. A. P., & DaCunha, J. P. (2007). Nursing Leadership and Management: Theories, Process and Practice.

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